PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview 3
2. Background & Business Objective 4
3. Product Purpose 6
4. System Functionalities 6
5. User Stories & Use Cases 10
6. Features Included (Scope Definition) 11
7. Features Not Included (Scope Control) 12
8. Assumptions & Constraints 12
9. Risks & Questions Throughout Development 13

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1. Project Overview

- **Problem Statement:** Traditional hospitality marketing operates on a human-centric paradigm, prioritizing legacy SEO, review sentiment, and persuasive copy. However, a fundamental shift in travel commerce is underway: AI agents are increasingly acting as autonomous proxies for users — navigating complex trip planning and property selection. Unlike humans, these agentic systems require high-signal, structured, and machine-interpretable data to execute decisions. Because most hotel listings remain optimized exclusively for human consumption, a critical visibility gap has emerged. Consequently, high-quality properties face systemic omission from AI recommendations simply because their digital presence is "illegible" to the engines now controlling the discovery layer.
- **Target Domain:** The primary domain is **AI Systems & Agentic Workflow Automation**, focusing on the design and orchestration of multi-step reasoning pipelines for autonomous task execution. The application is **Hospitality & AI-Led Discovery**, specifically targeting the technical mechanisms through which hotels are identified, evaluated, and selected by emerging AI travel agents. The strategic focus is **AI Engine Optimization (AEO)**, shifting the industry paradigm from human-centric SEO to the high-signal, machine-readable data structures required by generative decision engines.
- **Proposed Solution Summary:** WeBoosta is a **multi-agent AI system** that leverages **Google Gemini 3.1 Pro** to automate the discovery and optimization of your hotel for the agentic economy, simulating AI decisions to instantly refine your digital presence for autonomous agents. The platform executes an intelligent **nine-stage agentic pipeline** — orchestrated via **LangGraph** — to identify gap signals such as price clarity and amenity matching, and to automatically regenerate structured, AI-friendly content optimized for extraction by generative

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engines. The system delivers a **quantifiable scoring framework** and a **measurable before/after delta**, allowing hospitality operators to benchmark their visibility across distinct traveller personas and verify tangible ROI.

2. Background & Business Objective

- **Background of the Problem:** The hospitality industry is facing a transformative shift as AI travel agents increasingly replace traditional human-led search, creating a critical visibility gap for hotels. While legacy marketing focuses on persuasive copy for humans, WeBoosta addresses the technical necessity of AI Engine Optimization (AEO) by structuring content for machine readability. This multi-agent AI system leverages the **Google Gemini 3.1 Pro** reasoning engine — accessed via LangChain — to simulate agentic decision-making, allowing operators to understand why their properties are omitted from AI recommendations. By utilizing tool-augmented agents — including **Firecrawl** for live deep-scraping and **Tavily** for competitive AI-powered search — the platform replaces "black-box" guessing with real-world grounding. As 40% of global travellers migrate toward AI-led planning (Phocuswright, 2025), hotels failing to optimize for machine-readable signals face systemic displacement. WeBoosta provides a measurable before/after delta through a diagnostic "Discover Why?" workflow, ensuring properties are not only present but interpretable and trusted by generative engines. With the AI tourism market projected to reach USD 13.38 billion by 2030 (MarketsandMarkets, 2024), this solution offers a first-mover advantage by automating the creation of Schema.org markup and entity-driven content, ultimately reducing reliance on high-commission Online Travel Agencies.
- **Importance of Solving This Issue:** Solving this issue is paramount because it **bridges the systemic gap between human-centric marketing and machine-led discovery**, preventing high-quality hotels from becoming obsolete in an increasingly automated marketplace. By **automating the transition to AI Engine Optimization (AEO)**, the platform enables hospitality operators to bypass the **prohibitive costs and technical barriers** associated with manual data structuring.
- This ensures that properties can reclaim economic sovereignty by optimizing for direct AI agent discovery, effectively circumventing the 15-25% commission fees typically charged by legacy

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Online Travel Agencies (Financial Times, 2025). Furthermore, by providing a quantifiable before/after scoring framework, the solution transforms vague marketing efforts into auditable, data-driven strategies, allowing even small-scale operators to compete for visibility on a global stage without requiring years of technical SEO expertise. Ultimately, solving this problem future-proofs the hospitality industry, ensuring that a hotel's merit is accurately interpreted and prioritized by the digital gatekeepers of 2026.

- **Strategic Fit / Impact:** WeBoosta is purpose-built to align with the AI Systems & Agentic Workflow Automation domain of UMHackathon 2026 by leveraging **Google Gemini 3.1 Pro** as its central reasoning orchestrator, interfaced via LangChain and orchestrated through a **LangGraph** stateful graph. Rather than a simple linear chain, the system establishes a sophisticated multi-agent pipeline that manages stateful transitions across nine specialized nodes, including live tool-augmentation via Tavily AI search and Firecrawl deep-scraping. This architecture demonstrates the non-substitutable role of the reasoning engine; without Gemini 3.1 Pro's ability to execute complex diagnostic reasoning and synthesize unstructured data into machine-readable schemas, the system would lose its capacity to simulate the nuanced selection processes of autonomous AI agents. Real-time agent status updates are streamed to the frontend via WebSockets, enabling a live, transparent orchestration experience. By integrating a **Supabase/PostgreSQL** persistence layer for history tracking and offering export-ready optimized metadata, the solution facilitates a complete AI-native development lifecycle, bridging the gap between raw data extraction and deployable machine intelligence.
- Beyond technical execution, the platform delivers a profound commercial impact by addressing the hospitality industry's critical visibility gap in the era of AI Engine Optimization (AEO). As 40% of global travellers migrate toward AI-led planning (Phocuswright, 2025), hotels risk systemic omission from recommendation engines if their digital presence remains "illegible" to machines. WeBoosta empowers hospitality operators to reclaim economic sovereignty by optimizing for direct AI agent discovery, effectively circumventing the 15-25% commission fees associated with legacy Online Travel Agencies (Financial Times, 2025). By providing a quantifiable before/after scoring framework and a diagnostic "Discover Why?" workflow, the system offers a defensible first-mover advantage, transforming passive marketing assets into high-signal data points that are both interpretable and trusted by the autonomous decision-engines of 2026.

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3. Product Purpose

3.1. Main Goal of the System:

- The primary objective of WeBoosta is to provide a comprehensive decision-intelligence platform that enables hospitality businesses to achieve maximum visibility within the emerging landscape of AI-led travel discovery. By bridging the structural gap between human-centric marketing and machine-readable data, the system ensures that high-quality hotels are not systematically omitted by the AI agents that now influence 40% of global travel planning (Phocuswright, 2025).

3.2. Intended Users (Target Audience):

- Hospitality Workers
- Hospitality Revenue & Property Managers
- Digital Marketing & AEO Specialists

4. System Functionalities

4.1. Description:

- The system operates as a stateful, tool-augmented agentic engine designed to bridge the structural visibility gap between traditional hotel marketing and the high-precision requirements of autonomous AI decision-makers. Powered by **Google Gemini 3.1 Pro** — accessed via LangChain and orchestrated through a LangGraph StateGraph — the platform coordinates a nine-stage multi-agent pipeline that captures nuanced traveller personas, competitive market signals, and machine-readable requirements to simulate and improve a property's likelihood of being selected by AI travel agents.
- Unlike static models, the intelligence is grounded in real-world data through integrated search and scraping tools: Tavily AI Search for deep hotel discovery and market research, Firecrawl for high-fidelity web scraping and content extraction, and DuckDuckGo for auxiliary information gathering. This tool-augmented architecture enables a diagnostic "Discover Why?" workflow that reveals exactly where a hotel's digital footprint fails to provide sufficient signals for AI extraction.

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- The system captures the specific logical weights of varied personas — including Budget, Luxury, Family, and Business — and automatically executes structural content regeneration to produce Schema.org markup and entity-driven copy specifically optimized for generative engines like ChatGPT and Google AI Overviews. All agent state transitions and progress updates are streamed to the frontend in real time via WebSockets, providing a transparent, live view of the pipeline executing. Finally, the engine validates improvement through a quantifiable Before/After Score Delta, ensuring every optimization is backed by an auditable, 5-criterion sub-score breakdown that proves a measurable increase in AI-readability and competitive ranking.

4.2. Key Functionalities:

- **Nine-Stage Agentic Pipeline (LangGraph):** Discovery & Intelligence → Web Research & Extraction → Data Aggregation → AI Decision Simulator → Gap Analyzer → Autonomous Optimization → QA Validation → AI Re-simulation → Human-in-the-Loop (HITL) Approval. Each stage is a discrete LangGraph node with defined inputs, outputs, and conditional transitions.
- **AI Decision Simulation:** Scoring hotel profiles (0-100) based on 5 sub-criteria: Price Clarity, Location Relevance, Amenity Match, Review Sentiment, and Audience Alignment — each weighted equally at 20%.
- **Automated Machine-Readable Content Generation:** One-click generation of optimized hotel descriptions, entity-dense metadata, and Schema.org/JSON-LD markup specifically designed for AI agent extraction.
- **Comparative Analytics (Before/After Delta):** A quantifiable improvement score displayed on the dashboard after re-simulation, proving measurable ROI.
- **Persona-Based Visibility Simulation:** Simulating hotel performance across distinct traveller query types — Budget, Luxury, Family, and Business — to surface segment-specific gap signals.
- **Real-Time WebSocket Streaming:** Live agent status updates streamed from the FastAPI backend to the Next.js frontend, providing users with a transparent, step-by-step view of the pipeline executing.
- **Interactive 3D Dashboard:** A visual interface built with React Three Fiber & Drei featuring a 3D rotating "Blackbox" representation of the agentic workflow and 5-criterion radar charts for investigating AI visibility.

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- **Supabase History & Export:** A Supabase/PostgreSQL persistence layer that archives before/after score deltas and optimization records, allowing users to track progress over time and export optimized signals for immediate deployment.

4.3. AI Model & Prompt Design

- **4.3.1. Model Selection:** Google Gemini 3.1 Pro was selected as the core reasoning engine for WeBoosta. Released in February 2026 as the next iteration in the Gemini 3 series, Gemini 3.1 Pro was specifically chosen for its state-of-the-art agentic capabilities: improved tool use, reliable multi-step execution, and a 1M token context window that makes it ideal for synthesizing diverse data points — such as hotel amenities, pricing contexts, and long-form customer reviews — into a singular, machine-readable profile (Google DeepMind, 2026).
- Gemini 3.1 Pro is accessed via LangChain's Google Generative AI integration and orchestrated through a LangGraph StateGraph. The model serves as a non-substitutable orchestrator; without its advanced reasoning and reliable structured JSON output capabilities, the system would lose the ability to perform the diagnostic reasoning required for AI Engine Optimization (AEO). The model is invoked through the Gemini API (gemini-3.1-pro-preview endpoint), with a dedicated gemini-3.1-pro-preview-customtools endpoint used for agent stages requiring precise custom tool invocation.
- **4.3.2. Prompting Strategy:** WeBoosta utilizes a multi-step agentic prompting strategy to orchestrate the transition of unstructured hotel data into high-signal, machine-readable assets. Rather than a single monolithic prompt, the system employs a nine-stage LangGraph pipeline where Gemini 3.1 Pro acts as the central reasoning engine managing stateful transitions between specialized nodes — including the Discovery Agent, Scraper Agent, Optimizer Agent, Validator Agent, and Re-simulation Agent. This approach is essential for the AI Systems & Agentic Workflow Automation domain, as it allows for modular debugging and ensures that the reasoning engine is only provided with the specific, tool-augmented context required for each discrete task.
- The selection of an agentic workflow is driven by the need for Chain-of-Thought (CoT) integrity and precise competitive grounding. By decomposing the diagnostic process, the system can first execute a "Discover Why?" simulation to identify exactly which logical signals — such as price

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clarity or location relevance — are missing from a property's digital footprint. This sequential logic ensures that the optimization phase is not merely a creative rewrite but a data-driven technical recovery, specifically designed to satisfy the extraction requirements of autonomous AI travel agents.

- Moreover, the multi-step strategy incorporates a validation and persistence loop that ensures the final output is technically sound and factually accurate. The Optimizer Agent generates structured Schema.org and JSON-LD markup, which is immediately parsed and verified by a Validator Agent to eliminate hallucinations or schema errors. By integrating a Human-in-the-Loop (HITL) checkpoint before the data is committed to the Supabase/PostgreSQL persistence layer, the system balances high-speed automated transformation with expert human oversight, providing a quantifiable Before/After Score Delta that proves a measurable increase in AI visibility.
- **4.3.3. Context & Input Handling:** The system manages unstructured data — such as raw HTML and extensive markdown content — through a specialized Scraper Agent that utilizes Firecrawl to deep-scrape property websites and extract high-fidelity signals. The platform leverages Gemini 3.1 Pro's 1M token context window, operating with a practical session limit of 128,000 tokens per property analysis to balance reasoning quality with inference cost. This is enforced via Pydantic schema validation in the FastAPI backend. When user inputs or scraped content exceed this threshold, the system implements a recursive summarization strategy, breaking the raw text into manageable chunks and synthesizing them into dense, high-signal summaries. This prioritized ingestion process ensures that critical decision-making factors — such as price clarity and amenity mentions — are preserved while redundant or low-value data is discarded.
- **4.3.4. Fallback & Failure Behavior:** The system manages model failures and hallucinations through a multi-layered resilience strategy combining automated technical checkpoints with human-in-the-loop oversight. Every response from Gemini 3.1 Pro is passed through a Response Parser that validates the output against a strict JSON schema via Pydantic; if the model fails to return a structured response or provides a malformed schema, the system triggers an automatic retry mechanism with an adjusted temperature setting to enforce higher determinism. To address the risk of hallucinations, the Validator Agent performs a cross-reference check between

the optimized content and the original scraped data from Firecrawl, assigning a Naturalness/Confidence score. If this score falls below a critical threshold, the system enters a graceful error state where the optimization is flagged for human escalation within the dashboard. For scraping failures caused by anti-bot protections, the architecture includes a mock data fallback to ensure demo reliability. This ensures that even if the underlying model or scraping layer deviates, the final data stored in Supabase remains factually grounded and high-signal.

5. User Stories & Use Cases

User Stories:

- **"As a Digital Marketing & AEO Specialist,** I want the system to automatically generate Schema.org markup and entity-driven copy so that my property's digital presence is specifically optimized for high-signal extraction by generative engines like ChatGPT and Google AI Overviews."
- **"As a Marketing Specialist,** I want to simulate my hotel's performance across distinct traveller personas (Budget, Luxury, Family, and Business) so that I can understand how different logical weights in AI decision-making affect our competitive ranking for specific market segments."
- **"As a Hospitality Property Manager,** I want to utilize the 'Discover Why?' workflow so that I can identify specific gap signals — such as price clarity or amenity matching — that are currently preventing my hotel from being recommended by AI travel agents."

Use Cases (Main Interactions):

- **Prompt to UI Flow:** Users enter a natural language traveller query (e.g., "Best luxury hotel for families in KL"), then WeBoosta parses the intent through the Discovery Agent, identifies the necessary competitive signal benchmarks via the Tavily API, and renders a real-time Competitive Ranking Dashboard — with live agent status streamed via WebSockets. This flow transforms a subjective user request into a structured visual representation of the "AI Visibility Gap", showing exactly where the property stands against live market rivals.
- **Optimization Flow:** The user selects a specific "Gap Signal" identified in the diagnostic dashboard and triggers the "Optimize Signal" action. WeBoosta then generates the target

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machine-readable Schema.org markup and entity-dense descriptions while strictly maintaining the property's **"Brand Voice Tokens"** (such as specific sentiment, tone, and key amenity highlights). This ensures that while the content is being "technicalized" for AI agents, it retains the unique identity and persuasive quality required for human conversion. The Validator Agent cross-references the output against original scraped data to eliminate hallucinations before the result is committed to Supabase.

6. Features Included (Scope Definition)

- **Interactive 3D Diagnostic Dashboard:** A visual interface built with React Three Fiber & Drei featuring a 3D rotating "Blackbox" representation of the agentic workflow and 5-criterion radar charts for investigating the AI visibility of a property.
- **Nine-Stage LangGraph Orchestration Pipeline:** A stateful multi-agent system powered by Gemini 3.1 Pro that automates the transition from raw web data to machine-readable signals via specialized Discovery, Scraper, Aggregation, Simulator, Gap Analyzer, Optimizer, Validator, Re-simulation, and HITL nodes.
- **Real-Time WebSocket Streaming:** Live agent status updates streamed from the FastAPI backend to the Next.js frontend via WebSockets, providing full transparency into the pipeline as it executes.
- **Automated Machine-Readable Content Generation:** One-click generation of optimized hotel descriptions, entity-dense metadata, and Schema.org/JSON-LD markup specifically designed for AI agent extraction.
- **Persona-Based Visibility Simulation:** Capability to test and rank hotel performance against live competitors across distinct traveller personas, including Budget, Luxury, Family, and Business.
- **Quantifiable ROI & History Tracking:** A persistence layer integrated with Supabase/PostgreSQL that records before/after score deltas and optimization history, allowing users to export optimized signals for immediate deployment.

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- **Human-in-the-Loop (HITL) Approval Node:** A mandatory verification step that allows users to review, edit, or override AI-generated optimizations — ensuring brand voice consistency and factual accuracy before finalization.

7. Features Not Included (Scope Control)

- **Real-time OTA Integration:** The current system does not provide direct, live API connections to legacy platforms like Booking.com or Agoda for real-time inventory or price management.
- **Production Environment Auto-Deployment:** While the platform generates optimized Schema.org markup and entity-dense copy, it does not automatically push these updates to a hotel's live production server or CMS.
- **Native Mobile Application:** The initial version is strictly a web-based dashboard built with Next.js; native iOS or Android applications are not included in the current development phase.
- **Simultaneous Multi-user Collaboration:** Support for multiple users accessing and editing the same hotel optimization session simultaneously is not supported in the MVP.
- **Automated Paid Ad Management:** The system focuses exclusively on organic visibility through AI Engine Optimization (AEO) and does not manage or optimize paid search (SEM) or social media advertising budgets.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The total token cost for an average user session is estimated at approximately 23,300 tokens, comprised of ~19,300 input tokens and ~4,000 output tokens distributed across the two-phase discovery process and the nine-stage agentic pipeline. Based on Gemini 3.1 Pro API pricing of approximately $3.50 per 1M input tokens and $10.50 per 1M output tokens, the calculated cost per session is approximately $0.11. To ensure these costs remain manageable at scale, 24-hour caching for repeated queries is implemented in the Supabase persistence layer. Additionally, the system uses fixed-length prompts and performs

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summarization, truncation, or chunking of oversized inputs to optimize context window usage and prevent unnecessary token expenditure.

- **Technical Constraints:** The platform operates with a practical session context limit of 128,000 tokens per property analysis to ensure Gemini 3.1 Pro operates within optimal reasoning bounds without incurring excessive latency or inference costs — well within the model's 1M token context window. The backend is built on Python 3.10+ with FastAPI for asynchronous API performance and Pydantic for robust data validation and schema enforcement. While the system is tool-augmented via Firecrawl and Tavily, scraping capability is limited by external site anti-bot protections; therefore, the architecture includes a structured mock data fallback to ensure reliability during demonstrations.
- **Performance Constraints:** Due to the sequential nature of the nine-stage pipeline, the system faces an inherent performance trade-off with an expected processing latency of 10 to 30 seconds per full run. To mitigate user drop-off, the dashboard includes real-time WebSocket progress indicators and asynchronous processing via FastAPI's async endpoints. The platform operates under the assumption that users possess stable internet connections and that hotel operators have access to standard devices for dashboard monitoring. The platform is not designed for full autonomy; all AI-generated optimizations must be reviewed and approved by a human expert via the HITL node before they are considered ready for deployment.
- **User Input:** The system is fundamentally dependent on the integrity of the initial hotel URL and traveller query, as aggressive anti-scraping protections can result in the Scraper Agent ingesting incomplete signals which risks skewing the final visibility score. To address the threat of AI hallucinations — particularly regarding malformed JSON-LD markup or inaccurate amenity extraction from unstructured HTML — the platform utilizes a multi-agent corroboration loop where the Validator Agent cross-checks every machine-readable output against the original raw data logs provided by Firecrawl. Because Gemini 3.1 Pro functions as a reasoning engine rather than a fully autonomous auditor, a mandatory Human-in-the-Loop (HITL) checkpoint is required before data is committed to Supabase/PostgreSQL, ensuring factual grounding and brand alignment in all final outputs.

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9. Risks & Questions Throughout Development

- **Optimization Uniqueness:** To ensure that the AI does not generate generic or "cookie-cutter" descriptions for different properties, the Discovery Agent actively incorporates real-time competitive data from Tavily and unique property signals (USPs) into the reasoning pipeline. By grounding the generation in the specific "signal gaps" identified during the competitive search phase, Gemini 3.1 Pro focuses on market-gap exploitation rather than broad hospitality templates, ensuring that the optimized output for every hotel remains distinct and highly competitive within its specific local market.
- **Markup Integrity:** The system mitigates the risk of technical breakage through the Validator Agent, which functions as a programmatic gatekeeper for all generated code. This agent performs a schema-level verification on all JSON-LD and HTML markup — enforced via Pydantic — to ensure it adheres to current Schema.org 2026 standards before it reaches the persistence layer. If the model produces malformed or broken code, the system triggers an automatic retry mechanism with higher determinism or enters a graceful error state, ensuring the final exported asset is production-ready.
- **Unstructured Data Contextualization:** When the system encounters messy HTML or ambiguous unstructured web data from Firecrawl, it utilizes a multi-agent corroboration loop to prevent hallucinations. The Aggregation Agent cross-references its findings against the raw scraping logs, and if a critical piece of information — such as a specific amenity or price policy — cannot be contextualized with a high confidence score, the system flags the signal for Human-in-the-Loop (HITL) review. This ensures that the AI does not guess or misinterpret the property's digital footprint, maintaining the factual integrity of the machine-readable profile stored in Supabase/PostgreSQL.
- **Scraping Reliability:** Hotels with aggressive anti-bot protections or dynamically rendered JavaScript pages may result in partial or failed Firecrawl extractions. This risk is mitigated through the structured mock data fallback, which activates automatically when the Scraper Agent detects an incomplete signal set. Users are notified of fallback activation via the dashboard, and mock data profiles are clearly marked to prevent confusion during demonstrations.

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